

1.0 Cellular Network Fundamentals & Frequency Reuse

In cellular communication systems, frequency reuse is the design technique of using the same radio frequencies in different geographic coverage areas (cells) separated by a minimum reuse distance D . A cluster of N cells shares the total available bandwidth $S = k * N$. Co-channel cells are cells that operate on the exact same set of radio frequencies. Co-channel interference (CCI) is minimized when the signal-to-interference ratio (SIR) exceeds the threshold of 18 dB.

2.0 Handoff Strategies & Cell Splitting Techniques

When a mobile station moves between adjacent base stations, seamless handoff algorithms prevent dropped calls. Hysteresis margins of 6 dB to 8 dB prevent ping-pong switching between base stations. To handle high user density in urban cores, network operators deploy cell splitting and microcells to multiply system capacity.